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Actionable Interpretability for Churn Classification: A Text Bottleneck Model Case Study at a Major Telecom Provider

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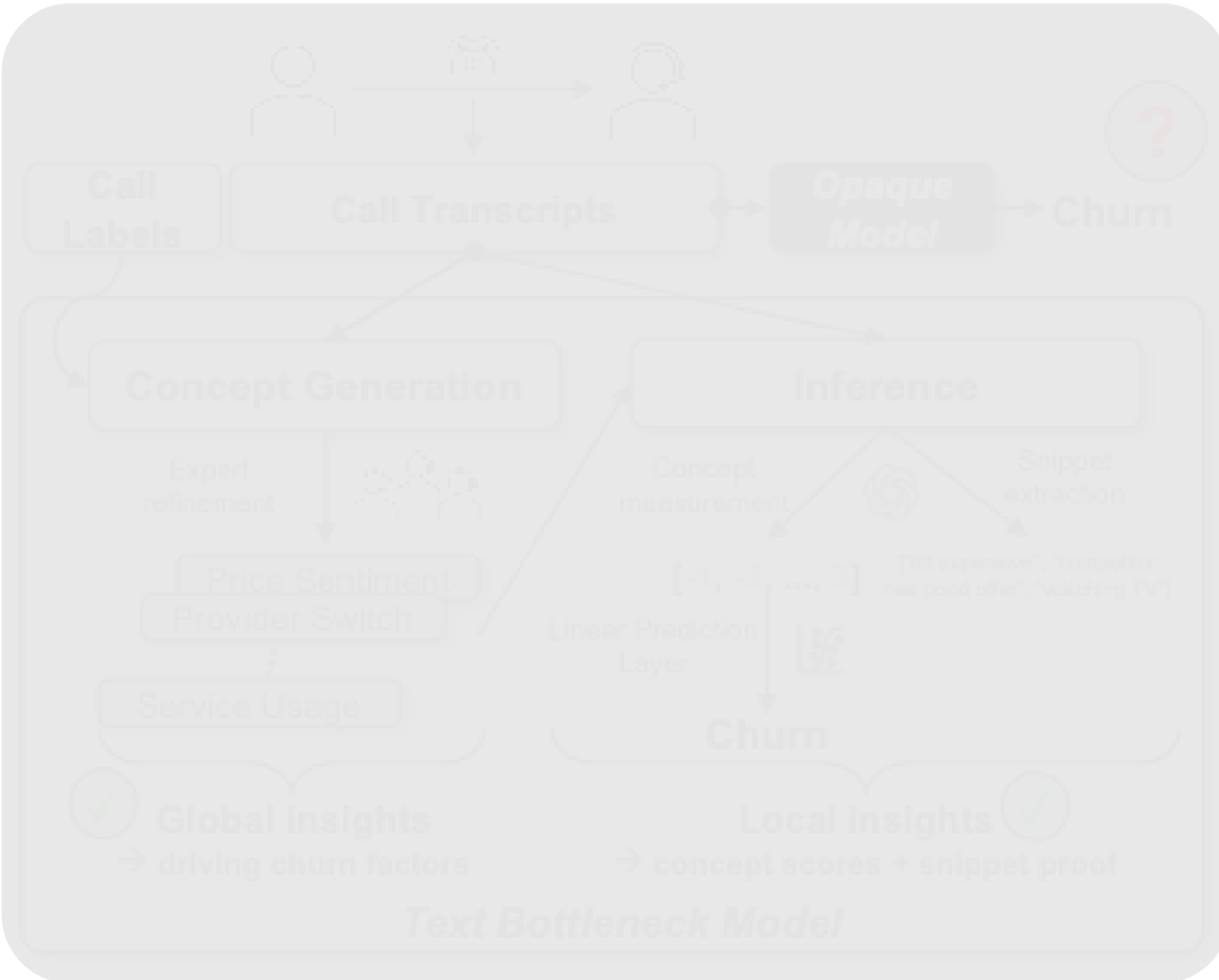
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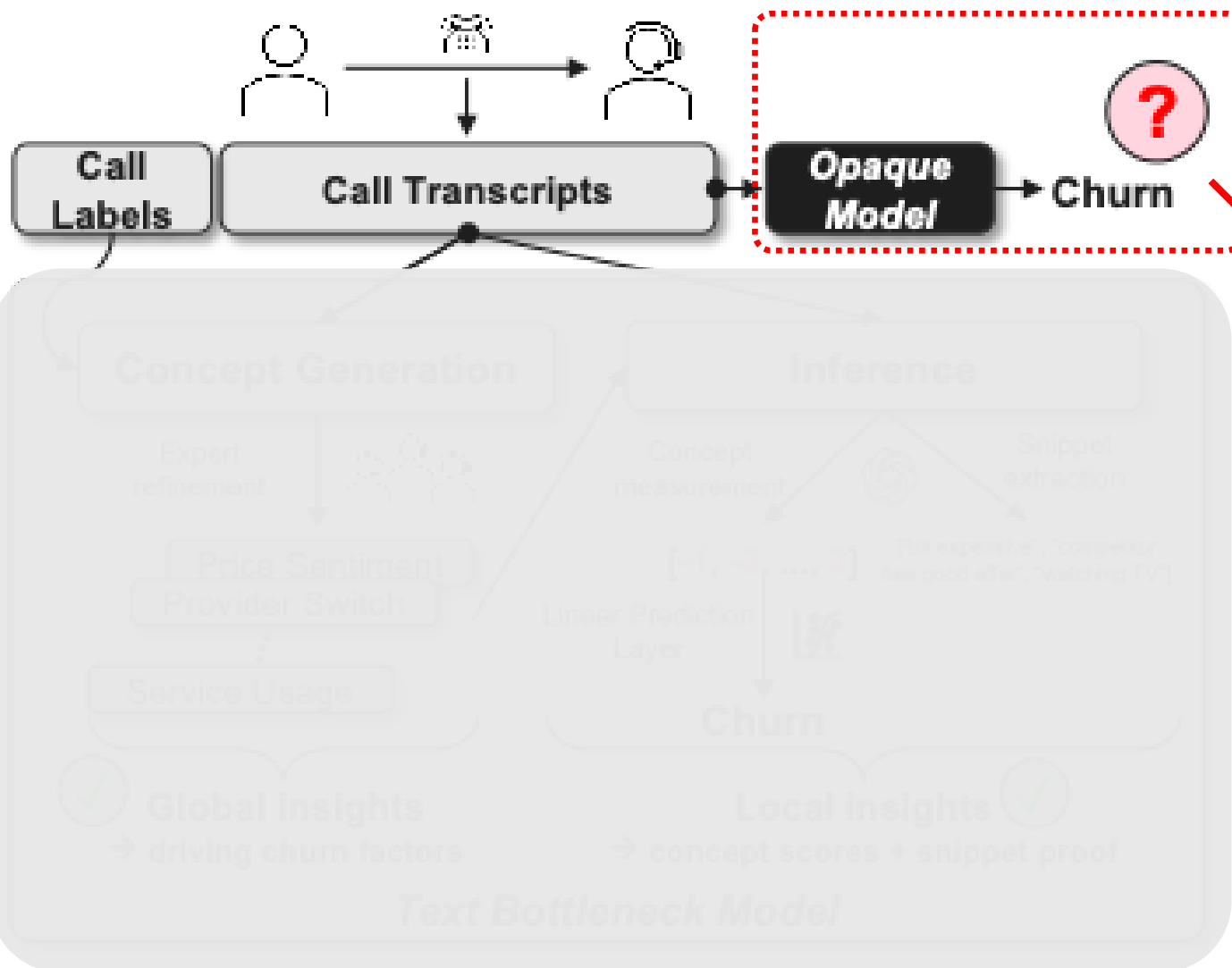
Background & Motivation

Context

- Managing customer churn is vital for subscription-based businesses



Background & Motivation



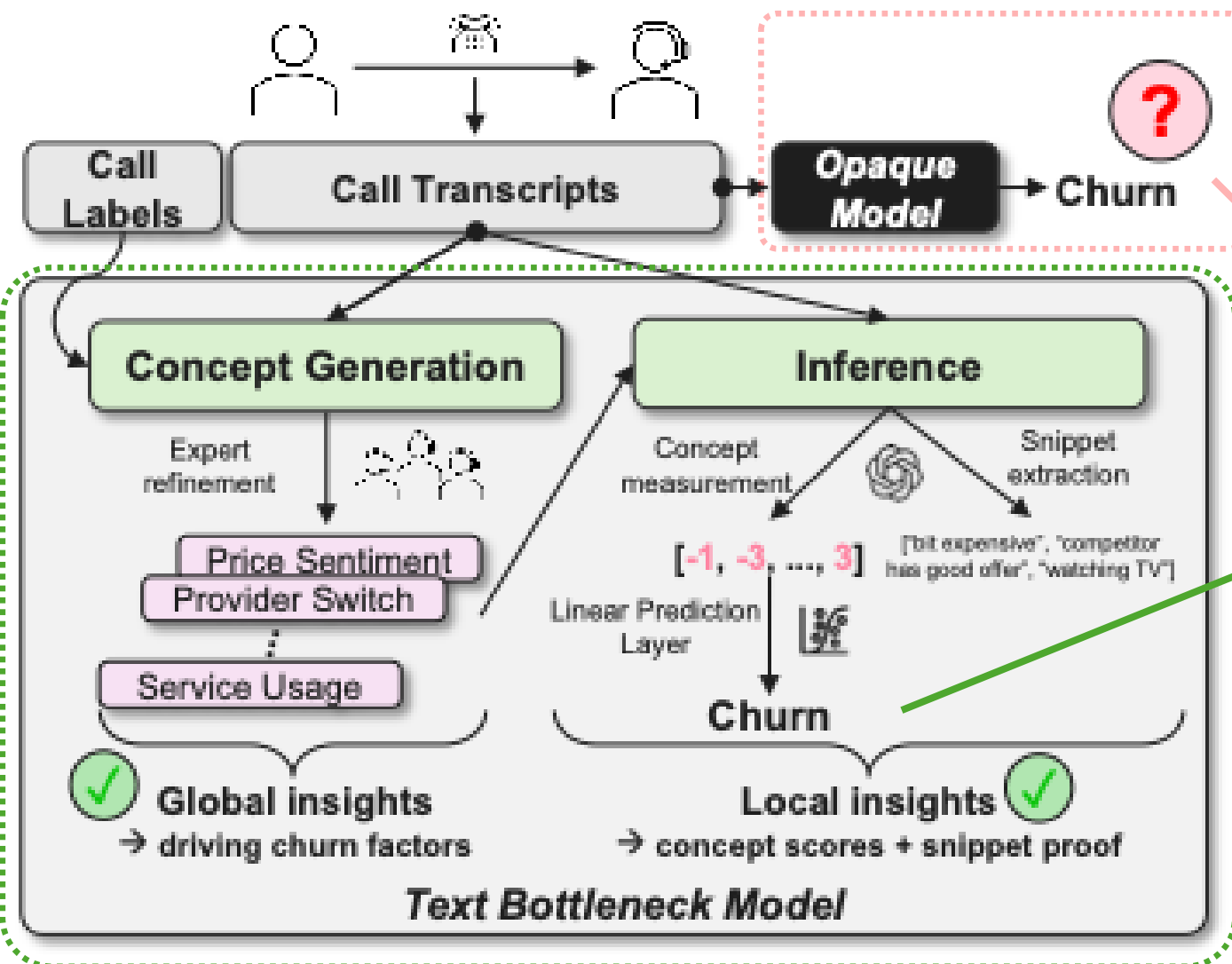
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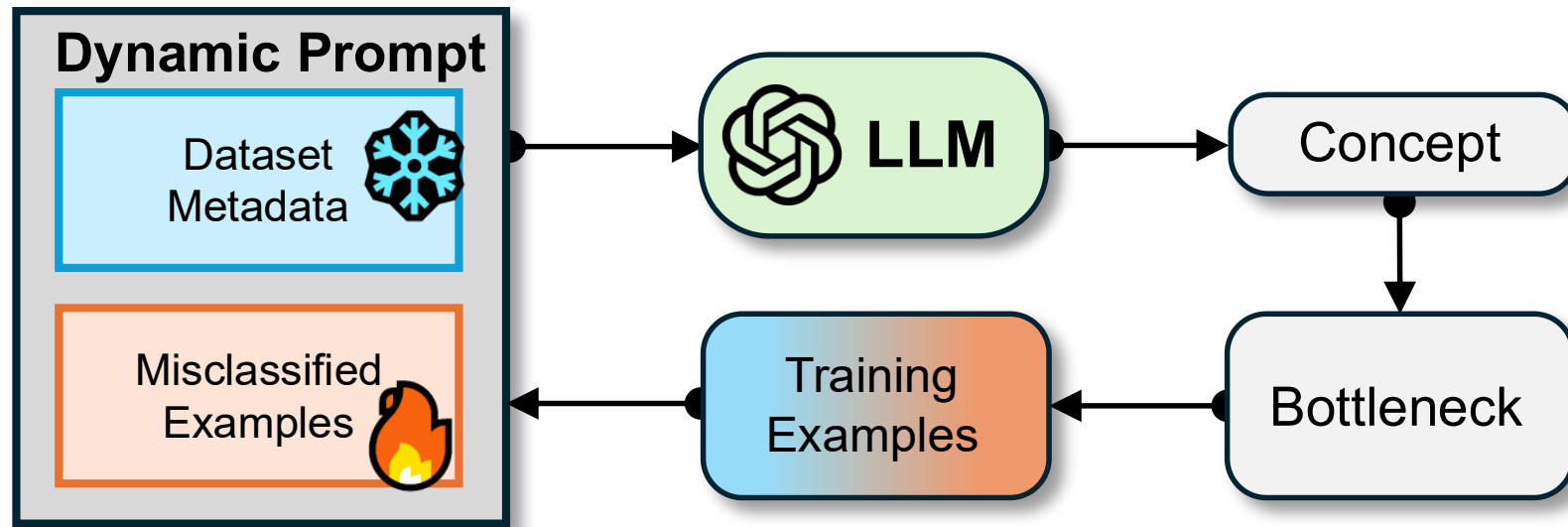
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Our solution

Text Bottleneck Model (TBM) [1]: interpretable concept layer → actionable retention guidance

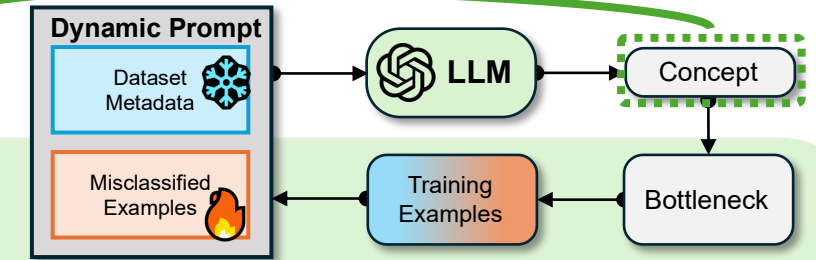
Text Bottleneck Model – Concept Generation



LLM iteratively proposes concepts targeting previously misclassified examples

Text Bottleneck Model – Concept Example

Relocation Mention



- **Description:** Relocation Mention captures whether the customer mentions moving to a new home, and whether this move is associated with continuing or canceling [TelCo] service.
- **Question:** Does the customer mention moving houses, and do they express an intent to cancel or continue their [TelCo] service as a result?
- **Response Guide:**
 - *Moving and Canceling Service (-3):* The customer clearly states that they are moving and plan to cancel their [TelCo] service or taking a subscription from a different provider at the new address.
 - *Possible Relocation Mentioned (-1):* There is an indirect mention or hint of a move, without confirmation or clarity on service continuation or cancellation.
 - *No Mention of Relocation (0):* There is no mention of moving houses or any change in living situation during the call.
 - *Moving and Continuing Service (3):* The customer mentions moving but also expresses an intent to keep using [TelCo], such as arranging a transfer of service.

Text Bottleneck Model – Concept Measurement

Given a transcript, an LLM is prompted with each concept definition

Concept: Relocation Mention

[agent]: Good afternoon, welcome to [TelCo], how can I help you?

[customer]: I'm **moving soon and would like to cancel my subscription**, because I'm not using it anymore.

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Response: Moving and canceling service Concept Score: -3

Responses are mapped to numerical scores

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Explanation: One snippet for **Moving and canceling service**. The customer mentions that they are moving and that they want to terminate the service.

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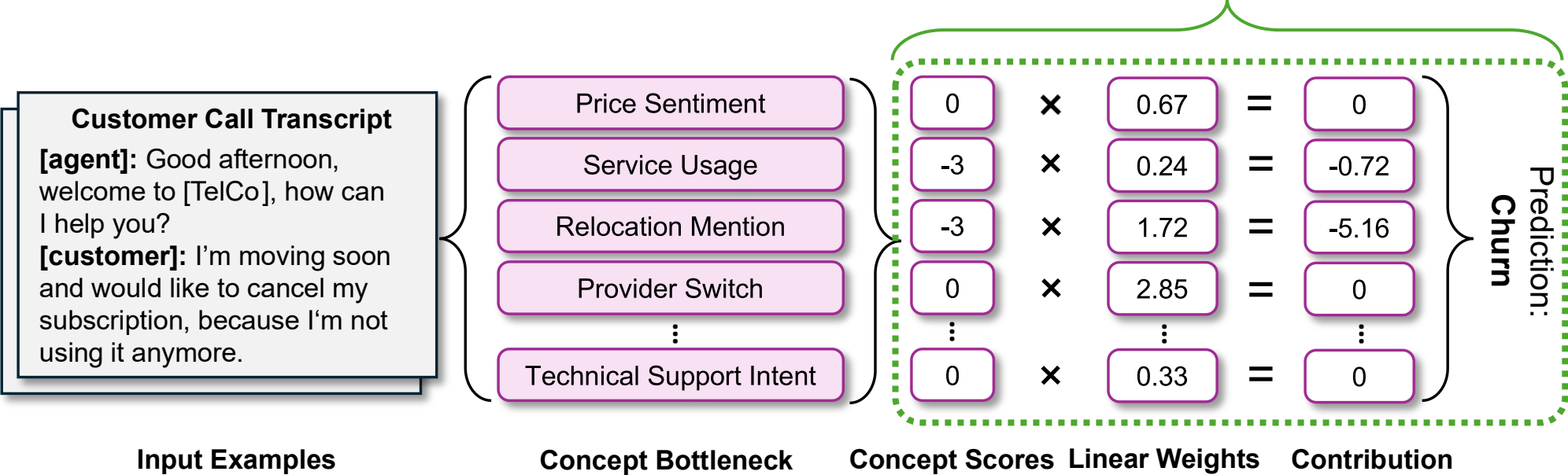
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→ **Local interpretability:** Extracted snippets and concept scores offer local explanations

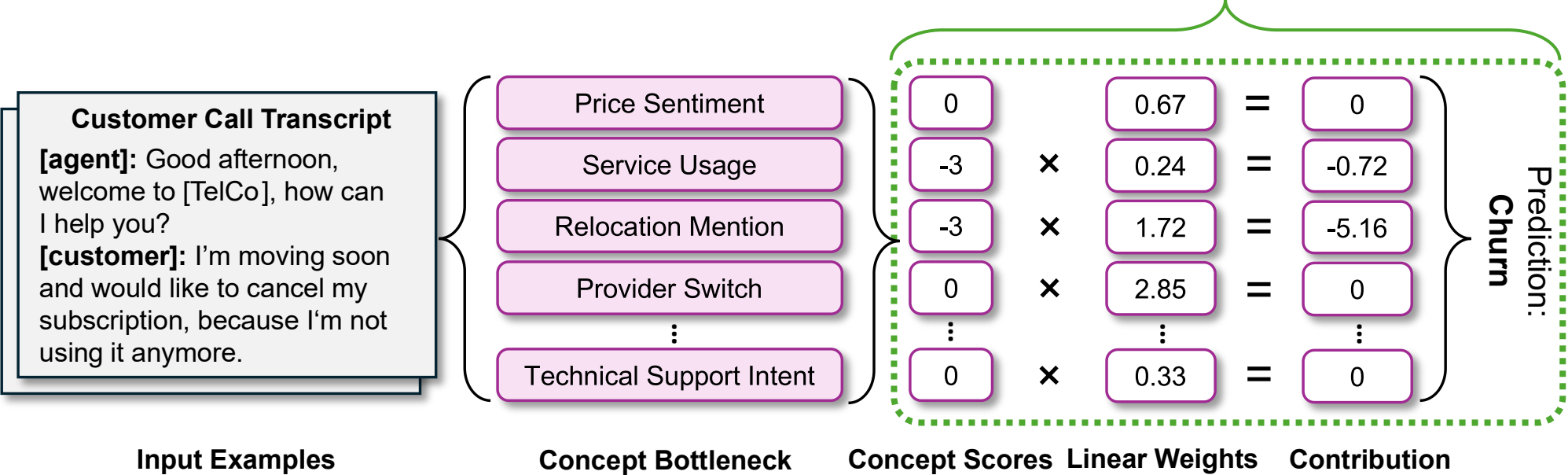
Text Bottleneck Model – Prediction Layer

White-box classifier takes vector of concept measurements and predicts target label



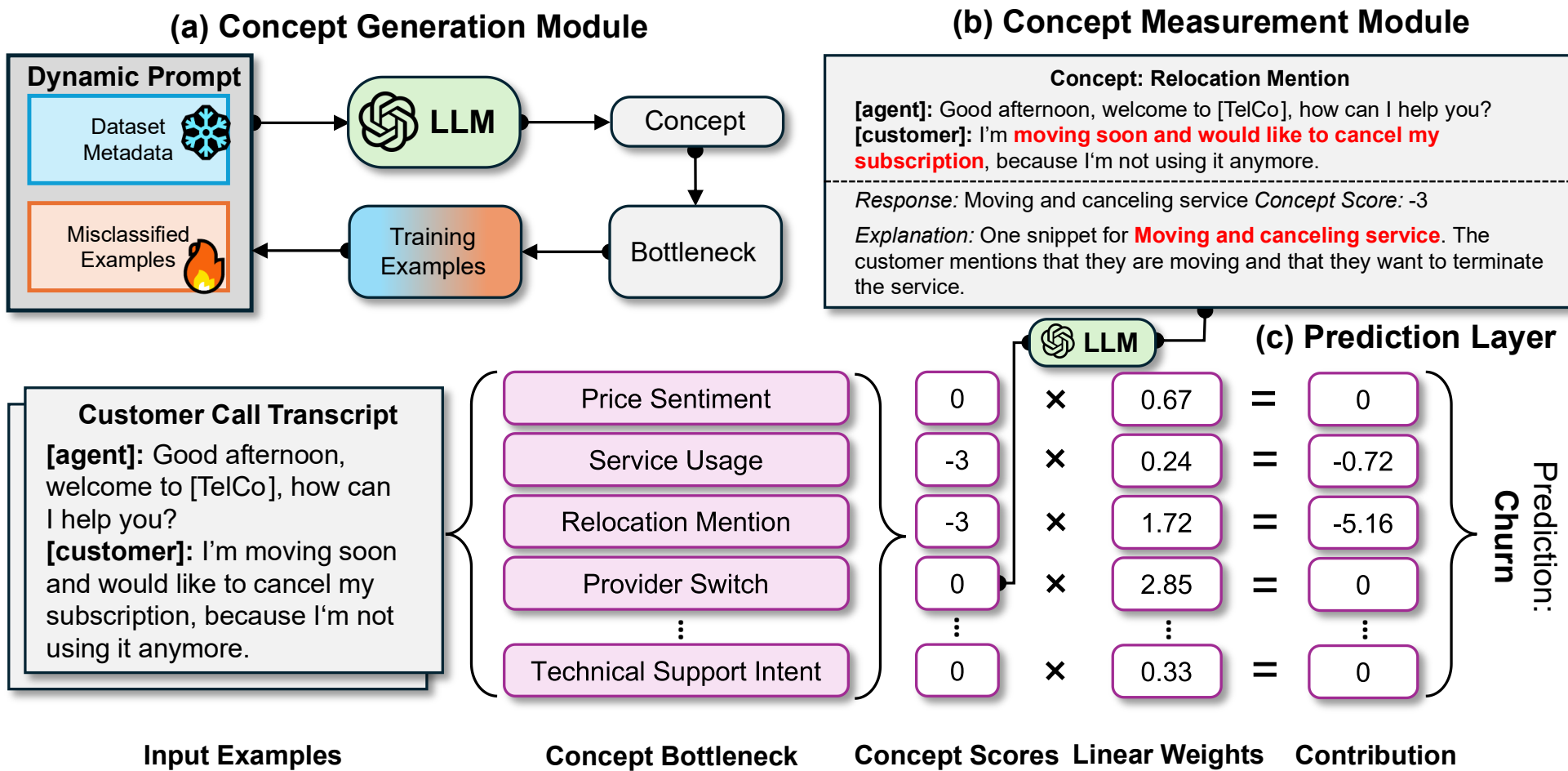
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→ **Global interpretability:** Learned weights indicate concept importance across dataset

Text Bottleneck Model – Overview



Experimental Setup – Dataset & Models

Dataset

- Expert-annotated dataset in Dutch
- Artificially balanced **train set** (100 Churn, 100 No Churn samples)
- „Natural“ **test set**: 28 Churn, 172 No Churn samples



Models

- TBM: **gpt-4-turbo-128k**, gpt-4o-mini
- Baseline (**black-box**): Domain-adapted BERTje [2], gpt-4-turbo-128k, gpt-4o-mini



Results – Churn Prediction

Framework	Prediction Model	F1-Score	Cohen's κ	AUPRC
Black-Box Production Model	BERTje	0.9081	0.8429	–
Black-Box GPT-4	gpt-4-turbo-128k	0.9485	0.8996	–
TBM	Logistic Regression (no reg.)	0.9198	0.8655	0.9254
	Logistic Regression (L1-reg.)	0.9184	0.8621	0.9230
	Logistic + Interactions (no reg.)	0.8780	0.7984	0.8812
	Logistic + Interactions (L1-reg.)	0.8615	0.7752	0.8645
	Decision Tree	0.8835	0.8091	0.8522

Finding 1: TBM with 7 expert-refined concepts performs competitively with blackbox models

Finding 2: Expert-refined concepts outperform automatic concept generation (max. F1-score: 0.7325)

Results – Final Set of Concepts

stronger churn signal → weaker (log. Regression coefficients)

-2.85

Provider Switch

Customer mentions switching to or from a competitor

-1.77

Service Modification Intent

Customer inquires about downgrading or upgrading service

-1.72

Relocation Mention

Customer indicates they are moving and may cancel

-0.67

Price Sentiment

Customer mentions satisfaction/dissatisfaction with pricing

-0.61

Subscription Flexibility Concern

Customer mentions contract flexibility

-0.33

Technical Support Intent

Customer mentions tech issues and whether they seek help

-0.24

Service Usage

Customer mentions whether they have stopped using the service

Results – Qualitative Error Analysis

	Churn intent present	Churn intent absent
Concepts activated		
No concepts activated		

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Concepts activated	Correct prediction: Churn ✓ <i>"I'm moving and want to cancel my subscription"</i> → Relocation Mention: -3 → Churn	
No concepts activated		Correct prediction: No Churn ✓ <i>"I'd like to upgrade my plan"</i> → no churn concepts activate → No Churn

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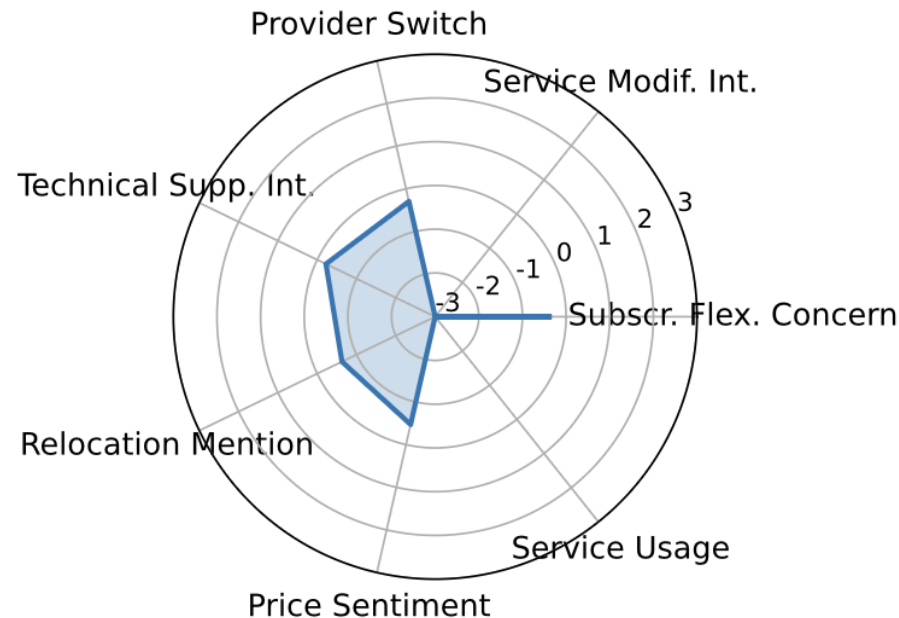
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Concepts activated	Correct prediction: Churn ✓ <i>"I'm moving and want to cancel my subscription"</i> → Relocation Mention: -3 → Churn	False positive: Churn ✗ <i>"The price is too high and the service is inflexible"</i> → Price Sentiment + Subscription Flexibility activate, but customer stays
No concepts activated	False negative: No Churn ✗ Concept scores don't cross threshold: <i>"I'm a bit unhappy with the price and the contract flexibility and want to switch"</i> → score too low No concept triggered: <i>"I just want to stop"</i> → no reason given	Correct prediction: No Churn ✓ <i>"I'd like to upgrade my plan"</i> → no churn concepts activate → No Churn

Results – Identifying Call Profiles

K-means in concept space to identify profiles (K=4)

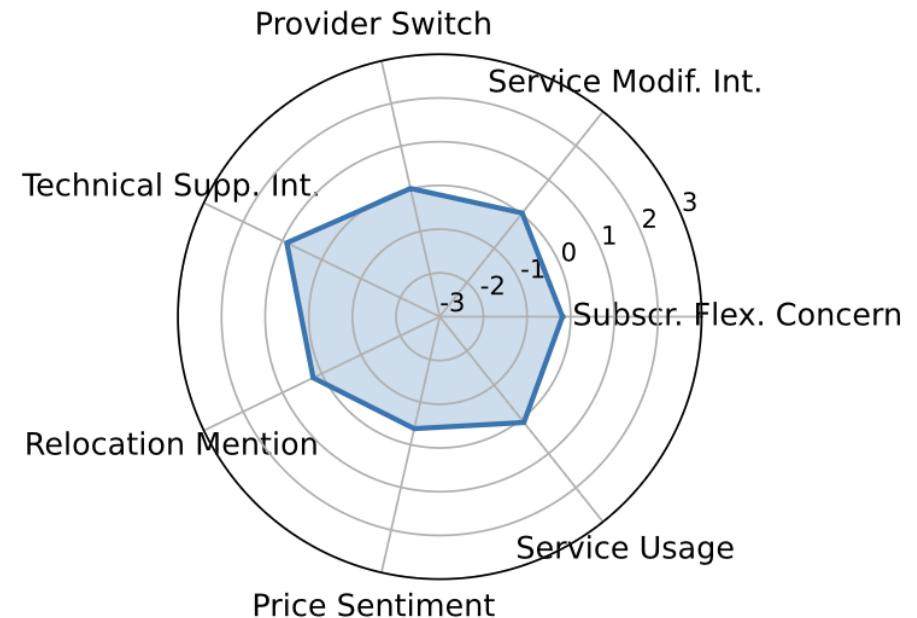
Cluster 2 Profile (108 samples)

Actual Churn Rate: 98.1% | Predicted Churn Rate: 99.1%



Cluster 4 Profile (128 samples)

Actual Churn Rate: 15.6% | Predicted Churn Rate: 7.8%



Finding 3: Concept space contains business-relevant information that is easily accessible

Results – Additional Results

Finding 4

0.943

avg. triple agreement

→ **Concept measurements are consistent**

Repeated concept scoring produces
stable results across runs

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Finding 5

88%

direct transcript matches

→ **Snippets are faithful and sufficient**

Independent LLM can predict concept scores from snippets alone (~90% accuracy) — no full transcript needed

Results – Additional Results

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Finding 5

88%

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→ **Snippets are faithful and sufficient**

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Finding 6

84.49%

F1 without chain-of-thought

→ **Chain-of-thought improves performance (91.98%)**

Removing the „thoughts“ section from the prompt measurably reduces F1-score

Interactive Dashboard

Concept Explorer for Call Transcripts

 About This Webapp

This dashboard uses **Concept Bottleneck Models** to make AI decisions transparent — breaking down customer service calls into key human-understandable concepts.

How it works: Enter a contact ID to fetch a transcript. An LLM scores key concepts and highlights the exact parts of the conversation behind each score. You can also learn more about the individual concepts by clicking on 'Show Concept Details'.

Churn Prediction: Once analyzed, you'll see two churn classifications: one from the production TECLA model, and one based on concept-driven reasoning.

 Search by Contact ID

JXCG5WTV6TZYSAQAO0F8U8DYLPEXKNY28BVB

Search

 Processing Complete!

Sample fetched and labeled successfully!



TECLA Classification: Churn

Concept Bottleneck Model Classification: Churn

Concept Progress:

Price Sentiment



Provider Switch



Relocation Mention



Service Modification Intent



Service Usage



Subscription Flexibility Concern



Technical Support Intent



Interactive Dashboard

Concept Measurement & Snippet Highlighting

Provider Switch

×

▼

Model Measurement:

competitor mentioned directly

Model Thoughts:

The customer explicitly mentions Ziggo and their offering as a reason for considering a switch away from KPN. They also state their intent to switch specifically to Ziggo, making this a clear case of 'competitor mentioned directly'. There is no indication of a switch to KPN or a lack of provider switch discussion.

Highlight Specific Options:

☒ competitor mentioned directly

☒ competitor implied

☐ no provider switch mentioned

☐ switch to KPN

Hide Concept Details

Transcript Viewer

[Agent]: Good afternoon, how can I help you?

[Customer]: Hi, I'm going to move within a couple of weeks and therefore I do not want to prolong my subscription. I want to check my options with you.

[Agent]: Of course, may I ask you why you don't want to prolong your subscription?

[Customer]: Actually I've been very happy so far with the prive and services of [Telco]. So it does not depend on those.

[Agent]: Nice to hear, what then makes you want to terminate your subscription?

[Customer]: With this relocation I prefer to have flexibility. I saw that [Company] has a move offer where I can get a shorter contract. With [TelCo], I'd be committed for a whole year and that does not with with my situation. Other providers are not so demanding.

Provider Switch

Provider switch is a situation where a customer explicitly mentions switching between providers, such as from Ziggo to KPN. This is a clear case of 'competitor mentioned directly'.

Example of a Provider Switch

Customer: I'm considering switching from Ziggo to KPN because they have a better offer. I want to check my options with you.

Agent: Of course, may I ask you why you don't want to prolong your subscription?

Customer: Actually I've been very happy so far with the prive and services of [Telco]. So it does not depend on those.

Agent: Nice to hear, what then makes you want to terminate your subscription?

Customer: With this relocation I prefer to have flexibility. I saw that [Company] has a move offer where I can get a shorter contract. With [TelCo], I'd be committed for a whole year and that does not with with my situation. Other providers are not so demanding.

Customer Guide

competitor mentioned directly

 The customer explicitly mentions switching between providers, such as from Ziggo to KPN. This is a clear case of 'competitor mentioned directly'.

competitor implied

 The customer mentions switching between providers, but does not explicitly mention the name of the provider they want to switch to. This is a case of 'competitor implied'.

no provider switch mentioned

 The customer does not mention switching between providers. This is a case of 'no provider switch mentioned'.

switch to KPN

 The customer explicitly mentions switching to KPN. This is a case of 'switch to KPN'.

Takeaways

Interpretability without compromise

TBM's match black-box performance while providing both global and local interpretability into why customers churn

Concepts unlock actionable insights

Concept space enables customer trend analysis, what-if scenarios, and targeted retention strategies

A general-purpose framework

TBM's apply to any classification or regression problem

Thank you for your attention! :)



Paper



Adrian Sauter

References

- [1] Josh Magnus Ludan, Qing Lyu, Yue Yang, Liam Dugan, Mark Yatskar, and Chris Callison-Burch. 2023. Interpretable-by-design text classification with iteratively generated concept bottleneck. *arXiv preprint arXiv:2310.19660*.
- [2] Wietse de Vries, Andreas van Cranenburgh, Arianna Bisazza, Tommaso Caselli, Gertjan van Noord, and Malvina Nissim. 2019. Bertje: A dutch bert model. *ArXiv, abs/1912.09582*.